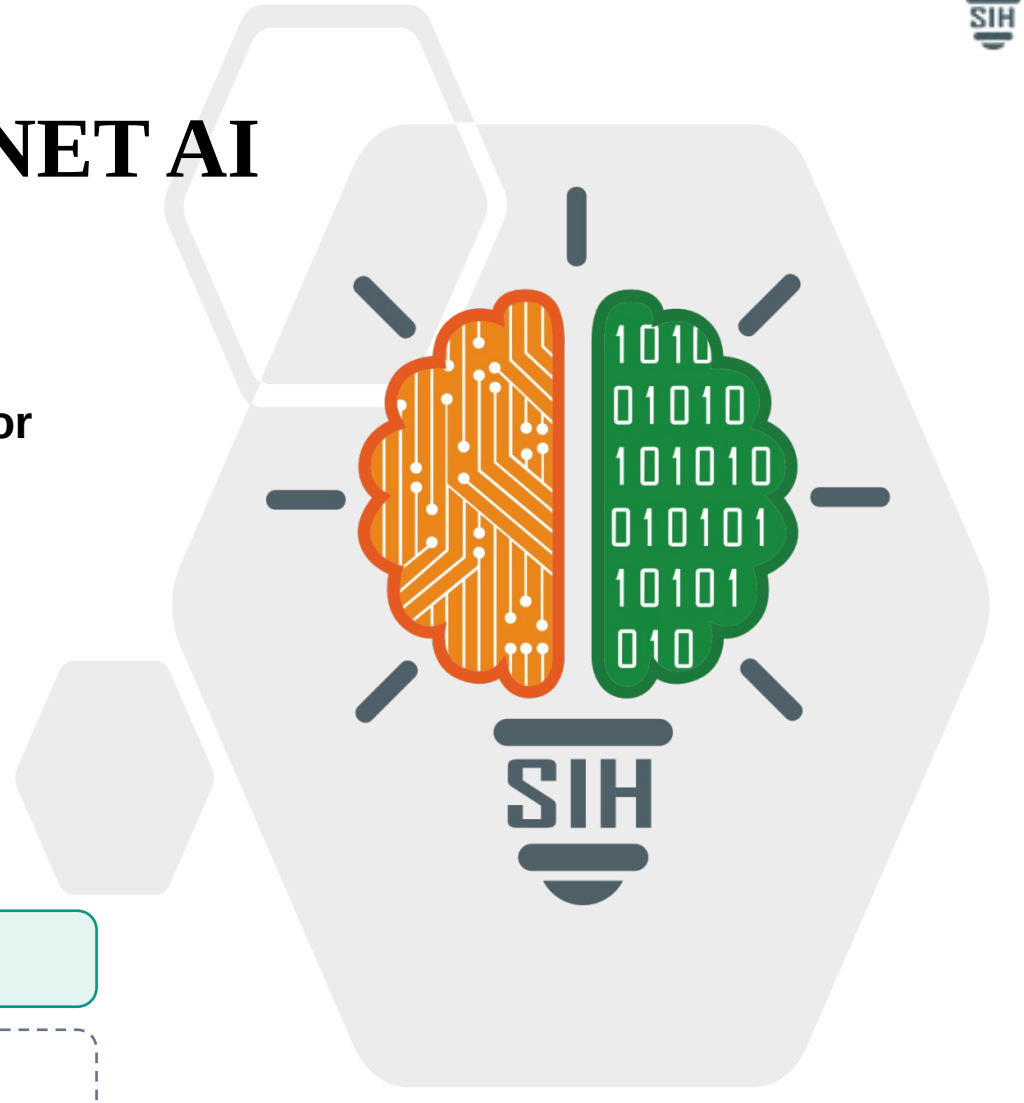
SMART INDIA HACKATHON 2026

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AEGISNET AI

Many vendor syntaxes. One security model. One auditable verdict.



- **Problem Statement ID – SIH26155**
- **Problem Statement Title – AI-Driven Multi-Vendor Network Security Compliance Auditor**
- **Theme – Blockchain and Cybersecurity**
- **PS Category – Software**
- **Team ID – 168312**
- **Team Name – Ansarash**

VENDOR COVERAGE

Cisco IOS ✓ Implemented	Juniper Junos ✓ Implemented
Fortinet FortiOS ✓ Implemented	Palo Alto PAN-OS ◇ Planned

Ansarash

AEGISNET AI

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❖ Proposed Solution

Unified multi-vendor security compliance auditor

1 INPUT

Cisco IOS, Juniper Junos or Fortinet FortiOS configuration, uploaded or pasted

▶

2 PARSE AND NORMALISE

Vendor detection → vendor adapter → Common Security Model with line-level evidence

▶

3 UNIFIED COMPLIANCE ENGINE

8 deterministic controls, severity-weighted score, NIST SP 800-53 and CERT-In mapping

▶

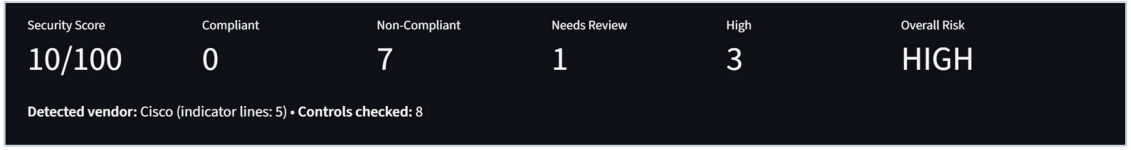
4 ACTIONABLE OUTPUT

Prioritised findings, vendor-specific fix with rollback, JSON report

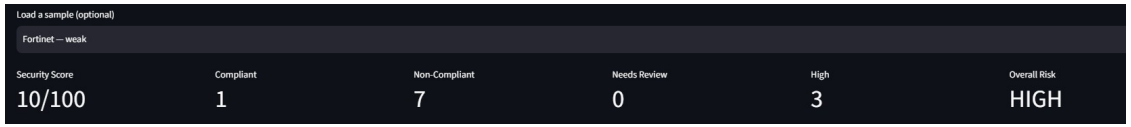
What makes AegisNet different

- ✓ Write a control once and audit it on every vendor
- ✓ Every finding points to the exact line in the original config
- ✓ Findings mapped to NIST SP 800-53 and the CERT-In Directions (2022)
- ✓ Fix and rollback commands generated per vendor
- ✓ Live re-audit: edit the config and watch the score change
- ✓ Secrets masked before anything is shown or exported
- ✓ A “Needs Review” state when a config can't prove a setting

Cisco IOS weak sample (live app)



Same engine, third vendor: Fortinet FortiOS weak sample (live app)



◇ **Planned** AI-assisted normalisation for unfamiliar syntax (with confidence score and human review), AI explanations of findings, a Palo Alto PAN-OS adapter.

AI interprets. Deterministic rules decide.

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TECHNICAL APPROACH

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Config › Vendor detection › Adapters (3 vendors) › Common Security Model › Rule engine › Findings + Remediation

Stack Python • Streamlit • line-based adapters (comments ignored) • rule table • JSON export ◇ LLM layer (Gemini, or a local model for air-gapped use)

✓ Add a vendor with one parser function ✓ Junos set + curly-brace, FortiOS config tree ✓ Unrecognised vendor halts the audit, no guessing

One control, every vendor

Common attribute	Cisco IOS	Juniper Junos	Fortinet FortiOS
`remote_logging = false`	no “logging host” line	no “set system syslog host” line	no syslogd server line

Finding AG-04 (Medium) Remote logging absent. Maps to NIST AU-2 and CERT-In Directions item (iv): logs retained for a rolling 180 days.

Vendor	Fix	Rollback
Cisco	`logging host <SYSLOG_IP>`	`no logging host <SYSLOG_IP>`
Juniper	`set system syslog host <SYSLOG_IP> any notice`	`delete system syslog host <SYSLOG_IP>`
Fortinet	`syslogd: set status enable; set server <SYSLOG_IP>`	`set status disable`

Cisco IOS Common Security Model (live app)

attribute	value	evidence
ssh_version	1	L2: ip ssh version 1
telnet_enabled	unspecified	—
http_mgmt_enabled	True	L3: ip http server
remote_logging	False	—
privileged_password	plaintext	L7: enable secret ********
snmp_default_community	True	L8: snmp-server community public RO
ntp_configured	False	—
login_banner	False	—

Juniper Junos (curly-brace) Common Security Model (live app)

attribute	value	evidence
ssh_version	2	L5: set system services ssh protocol-version v2
telnet_enabled	False	—
http_mgmt_enabled	False	—
remote_logging	True	L10: set system syslog host 10.1.1.5 any notice
privileged_password	missing	—
snmp_default_community	False	—
ntp_configured	True	L14: set system ntp server 10.1.1.1
login_banner	False	—

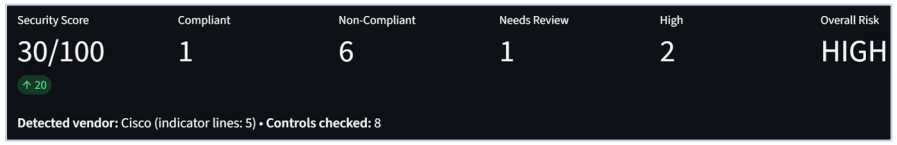
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FEASIBILITY AND VIABILITY

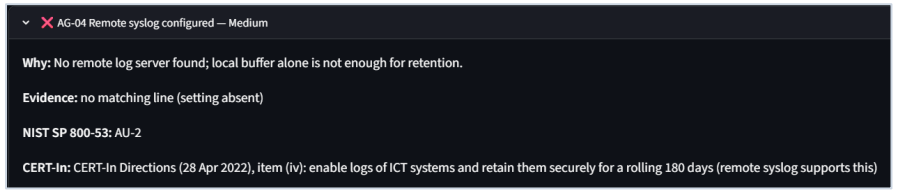
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01 Technically feasible: prototype status

Item	Status
Vendors	3: Cisco IOS, Juniper Junos (set + curly-brace), Fortinet FortiOS
Controls	8
Test configs	7 (6 valid, 1 invalid)
Weak Cisco	10/100
After fixing SSH	30/100 (+20)
Hardened Cisco	100/100
Deployment	Live on Streamlit Cloud



Live app: score after changing SSH to v2



Finding AG-04: evidence, NIST and CERT-In reference

02 Challenges and mitigation

- **Different syntax per vendor:** a separate adapter per vendor, one shared model
- **Version variations:** modular adapters; regression tests per version are planned
- **Incorrect AI mapping (planned):** confidence score, human review flag, AI never decides the verdict
- **Config can't prove a setting:** shown as “Needs Review”, not scored
- **Remediation risk:** every fix ships with a rollback; administrators review before applying

◇ AI guardrails (planned)

1. Secrets redacted before any AI step
2. AI maps unknown lines, with a confidence score
3. Low confidence flagged for human review
4. Rule engine makes the verdict

03 Scalability roadmap

◇ Planned

- Palo Alto PAN-OS adapter
- Firewall rule-base analysis (any-any, shadowed, redundant rules)
- CIS Benchmark mapping, plus NCIIPC and RBI
- Config drift tracking with a fleet-wide score
- PDF and SARIF reports

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IMPACT AND BENEFITS

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TARGET USERS

- Network administrators
- Cybersecurity teams
- MSPs
- Government and institutional networks
- Critical infrastructure operators

Security

Finds misconfigurations such as plaintext secrets, default SNMP communities, Telnet and HTTP management

Efficiency

One interface and one audit for multiple vendors; results appear instantly on the test configs

Consistency

The same control and the same score logic for every vendor

Explainability

Every finding shows its evidence line, framework reference, fix and rollback

Privacy

Configs are processed without any device access, and secrets are masked in output

India relevance

Checks aligned to the CERT-In Directions (2022) on log retention and NTP sync, relevant to government bodies, PSUs and MSPs. ◇ Planned: NCIIPC and RBI mappings.

KEY BENEFIT One unified, explainable security assessment across multi-vendor networks

Ansarash

RESEARCH AND REFERENCES

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Security frameworks

NIST SP 800-53 Rev. 5
AC-8, AC-17, AU-2, AU-8, CM-7, IA-5
✔ Used in prototype

CERT-In Directions, 28 April 2022
Under Section 70B(6) of the IT Act, 2000: items (i) and (iv)
✔ Used in prototype

CIS Benchmarks
Cisco IOS and Juniper Junos secure configuration
◇ Planned

DISA STIG
Security configuration guidance (reference)
Reference

NCIIPC guidelines and RBI Cyber Security Framework
India-specific mappings
◇ Planned

Vendor documentation

- ✔ Cisco IOS security configuration guides
- ✔ Juniper Junos security documentation
- ✔ Fortinet FortiOS administration and CLI documentation
- ◇ Palo Alto PAN-OS documentation

Technical research

- Multi-vendor configuration parsing
- Configuration normalisation
- Rule-based policy evaluation
- Explainable security analysis
- Vendor-specific remediation generation

Framework mappings show indicative alignment and are not a formal certification.